

BHARATHI S

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EDUCATION

B.E. Electronics and Communication Engineering, Chennai Institute of Technology (2023–2027)

CGPA: 8.9

EXPERIENCE

Software Development Intern

India Meteorological Department (Doppler Weather Radar) | Nov 2024

- Developed terrain-aware radar propagation models using Python, DSM/DEM datasets, and GIS tools. Modeled beam blockage and wave propagation to improve radar siting accuracy.
- Built geospatial workflows to process large scale terrain datasets for real-world meteorological decision systems.

Machine Learning Intern

Cognifyz Technologies | Aug 2024

- Built ML pipelines for prediction, clustering, and recommendation systems. Applied feature engineering and model tuning to improve performance.
- Developed visualizations for model insights and interpretability.

PROJECTS

API Reliability Lab (Production SaaS Platform) [Github Repo](#) | [Live demo](#)

Feb 2026

Next.js, TypeScript, PostgreSQL, Prisma, NextAuth, Redis, Docker, CI/CD

- Built a full-stack SaaS platform to simulate API failures (latency, rate limits, timeouts) and analyze backend reliability
- Implemented real-time streaming using SSE to deliver live latency metrics and simulation logs
- Optimized performance using Redis caching and PostgreSQL indexing, reducing repeated execution latency
- Designed secure multi-tenant architecture with NextAuth and Prisma, ensuring isolated user workflows
- Built scalable backend with fault-tolerant workflows and CI/CD using GitHub Actions and Docker.

Carbon Aware Serverless Scheduler [Github Repo](#)

Oct 2025

Google Cloud (Cloud Run, Functions, Firestore), Python, Docker, CI/CD

- Built a multi-cloud scheduler using ElectricityMap API to optimize workloads based on real-time carbon intensity and reduced estimated carbon emissions by ~87.8%.
- Designed serverless architecture using Cloud Run and Cloud Functions with automated CI/CD pipelines.
- Developed system monitoring and logging for distributed cloud workloads.

TaskFlow: Multi-Tenant Backend [Github Repo](#)

Mar 2025

FastAPI, PostgreSQL, Redis, JWT

- Designed a multi-tenant backend with secure JWT-based authentication, RBAC and data isolation. Built scalable REST APIs using FastAPI and PostgreSQL with data isolation across tenants.
- Optimized performance using Redis caching for read-heavy workloads.

SKILLS

Languages: Python, Java, C, SQL

Frontend: React.js, Next.js, TypeScript, JavaScript, HTML, CSS

Backend & Systems: REST APIs, API Design, Authentication (JWT, OAuth), RBAC, Data Pipelines, Caching (Redis), Scalable Systems

Databases: PostgreSQL, Redis, Firebase

Cloud/DevOps: GCP, Docker, CI/CD (GitHub Actions), Serverless Architecture

ML/AI: scikit-learn, Anomaly Detection, Model Deployment (MLOps), GenAI (LLM Integration)

CERTIFICATIONS

Google Cloud Certified: Cloud Digital Leader | Google Cloud Learning Paths: Cloud Engineering & Generative AI | SQL Programming: Certified by SkillRack and Coursera | CISCO Certifications: Programming in C, Networking & Cybersecurity Fundamentals

RESEARCH PUBLICATIONS

“A Literature Review on Cloud–DevOps Synergy for Scalable and Reliable Machine Learning Lifecycle Management”, IEEE International Conference on Smart Systems and Sustainability (ICSSS 2025) [Link](#)

Conducted a structured review of Cloud, DevOps, and MLOps architectures, identifying scalability, reliability and sustainability gaps in production ML systems. Proposed the Adaptive Cloud–DevOps Orchestration Framework (ACDOF).

ACHIEVEMENTS

Data structures and algorithms: Problem solved [Leetcode](#): 331, [code chef](#): 568, [skillrack](#): 745.